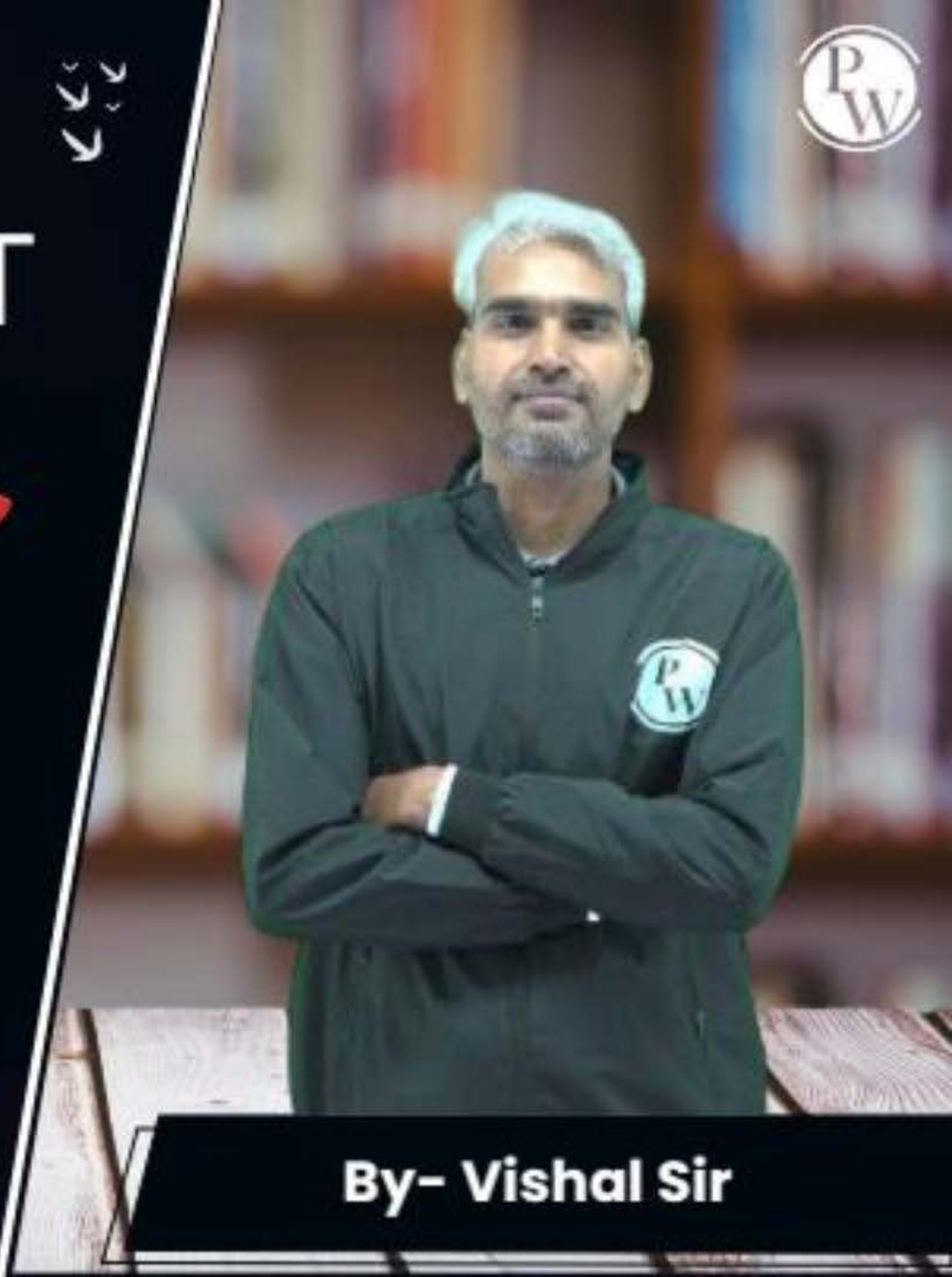
Computer Science & IT



Discrete Mathematics

Graph Theory

Lecture No. 08



By- Vishal Sir

Recap of Previous Lecture



Topic — Euler's equation for disconnected planar graph

Topic — Vertex coloring

Topic — Chromatic number

Slide

Topics to be Covered



Topic Vertex coloring

Topic Chromatic number

Topic Edge coloring

Topic Matching

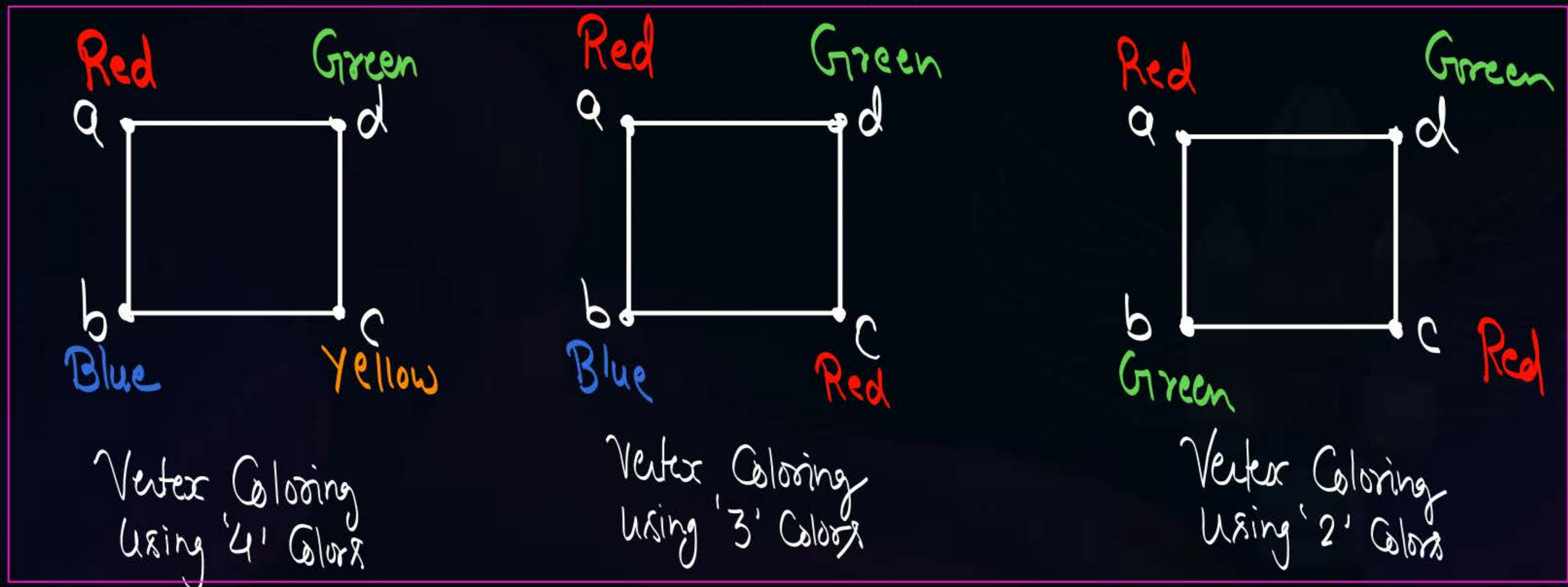
Slide



Topic : Vertex coloring

- An assignment of colors to the vertices of graph G, such that no two adjacent vertices of the graph have the same color is called vertex coloring of graph G.

All of them are valid Vertex Coloring



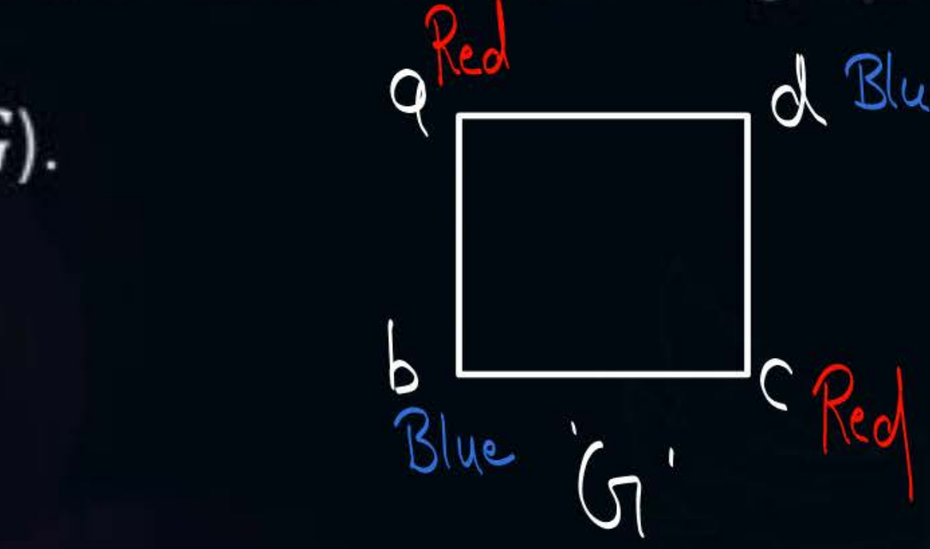
→ If we have a graph with 'n' vertices, then we can always Vertex Color that graph using 'n' Colors {i.e. one color per vertex}

"But we are interested in the minimum number of Colors required for the Vertex Coloring of the given graph"

Topic : Chromatic number

Minimum number of colors needed for the vertex coloring of graph G is called chromatic number of graph G,

It is denoted by $\chi(G)$.



We can not Vetex Color the graph G using less than '2' Colors

$\therefore \chi(G) = 2$



Topic : K-colorable vs K-chromatic

K-Colorable

* If there exist a vertex coloring of graph 'G' that uses 'K' colors, then graph G is called 'K' Colorable

If graph G is K-Colorable, then $\chi(G) \leq K$

K-Chromatic

If chromatic number of graph 'G' is 'K', then graph 'G' is said to be K-chromatic

If graph G is K-chromatic, then $\chi(G) = K$

Note:- ★

① If graph G is K-Colorable, then graph G is (K+1)-Colorable as well,

Provided $(K+1) \leq$ No. of vertices in graph G.

② If graph G is K-Colorable, then graph G may or may not be (K-1)-Colorable

③ If graph G is K-chromatic, then it is neither (K+1)-chromatic nor (K-1)-chromatic

Note:- ★

④ If graph G is k-chromatic then graph G is k-Colorable as well, but

if graph is k-Colorable then it need not be k-Chromatic



Topic : Four-Color Theorem

Every planar graph G is four colorable.

For any planar graph G, $\chi(G) \leq 4$

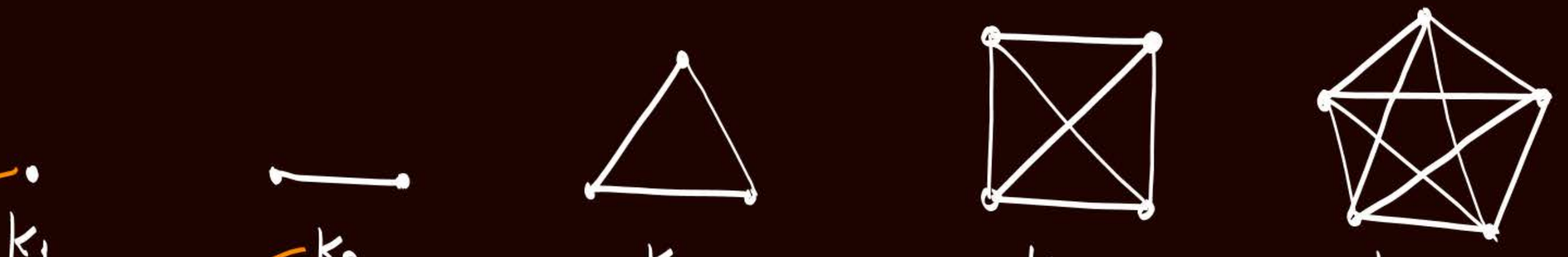
Note :→

① Chromatic number of graph G Can be '1' if and only if graph 'G' is a NULL graph

i.e. $(\chi(G)=1) \iff$ 'G' is NULL graph

② If graph G is not a NULL graph then $\chi(G) \geq 2$

Q:- Find the Chromatic number of Complete graph 'K_n'



K_1 K_2 K_3 K_4 K_5 . . .

$\chi(K_1) = 1$ $\chi(K_2) = 2$ $\chi(K_3) = 3$ $\chi(K_4) = 4$ $\chi(K_5) = 5$

For a Complete graph K_n, $\chi(K_n) = n$

Max degree

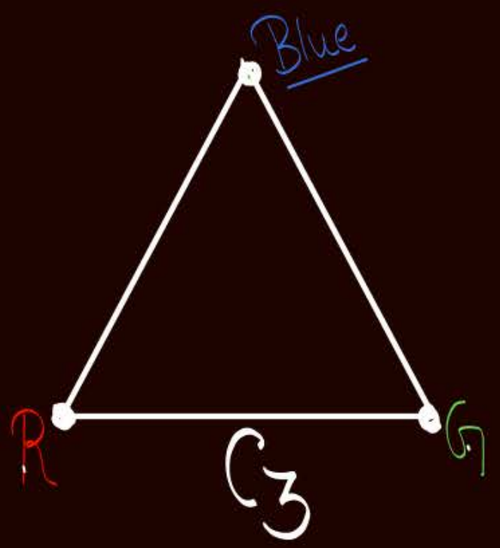
$\Delta(K_1) = 0$ $\Delta(K_2) = 1$ $\Delta(K_3) = 2$ $\Delta(K_4) = 3$ $\Delta(K_5) = 4$

$\chi(K_1) = 1 + \Delta(K_1)$ $\chi(K_2) = 1 + \Delta(K_2)$ $\chi(K_3) = 1 + \Delta(K_3)$ $\chi(K_4) = 1 + \Delta(K_4)$ $\chi(K_5) = 1 + \Delta(K_5)$

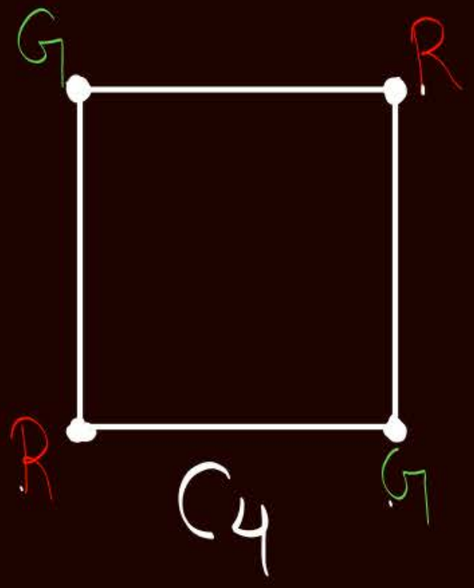
For any graph G,

$$\chi(G) \leq 1 + \Delta(G)$$

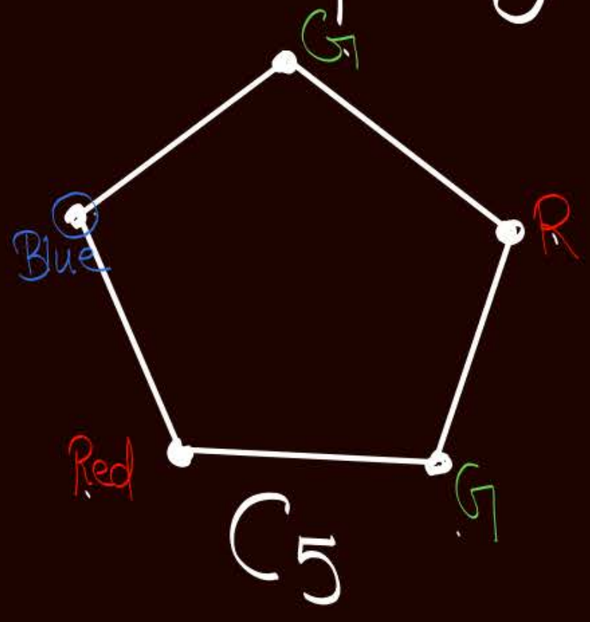
Q:- Find the Chromatic number of Cycle graph 'Cn'



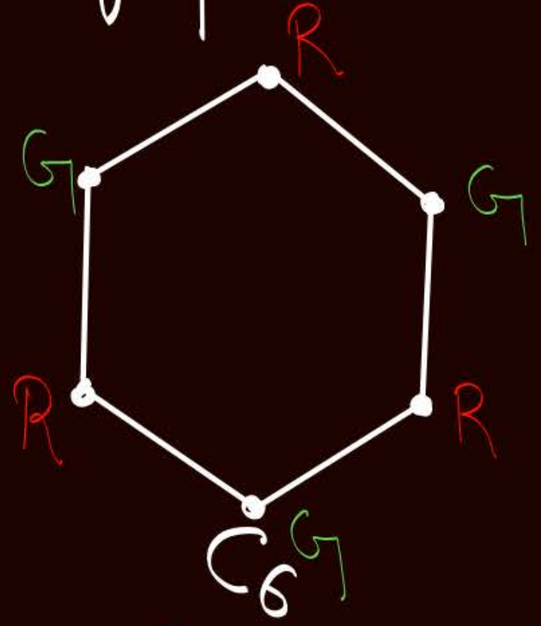
$\chi(C_3) = 3$



$\chi(C_4) = 2$



$\chi(C_5) = 3$

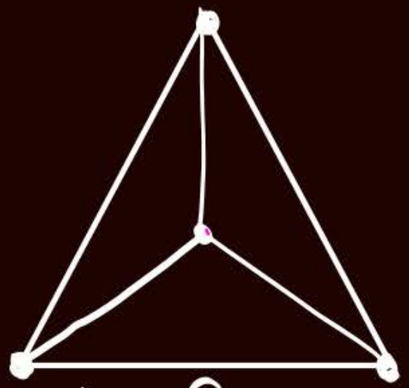


$\chi(C_6) = 2$

$$\chi(C_n) = \begin{cases} 2, & \text{if } n = \text{Even} \\ 3, & \text{if } n = \text{odd} \end{cases}$$

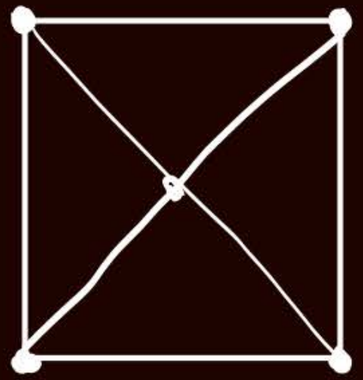
= it is also $(1 + \Delta(G))$

Q:- Find the Chromatic number of Wheel graph W_n



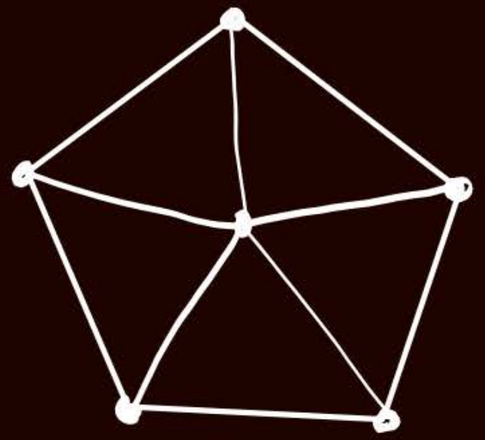
$W_4 = C_3 + 1 \text{ hub}$

$\chi(W_4) = 3 + 1 = 4$



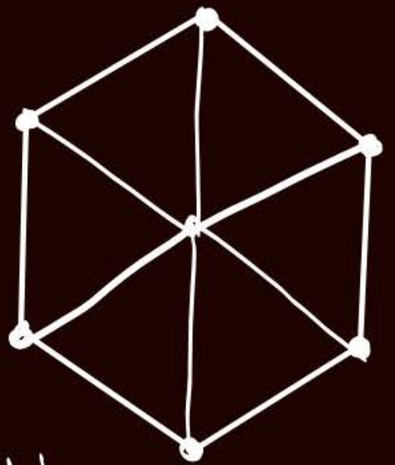
$W_5 = C_4 + 1 \text{ hub}$

$\chi(W_5) = 2 + 1 = 3$



$W_6 = C_5 + 1 \text{ hub}$

$\chi(W_6) = 3 + 1 = 4$



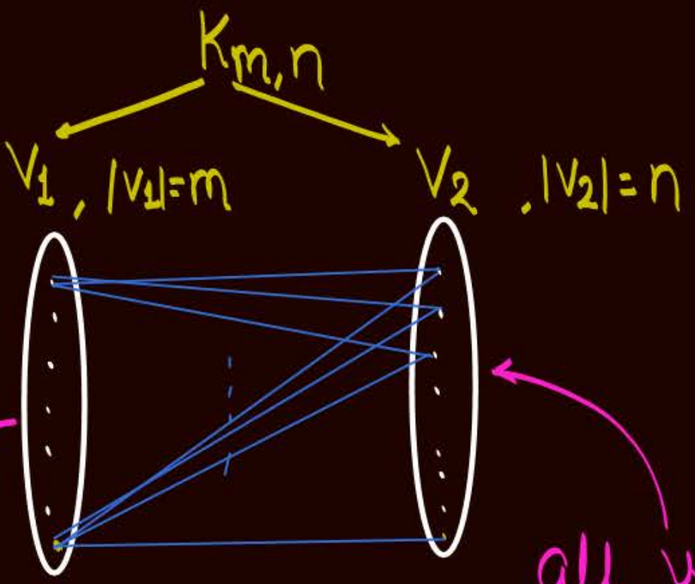
$W_7 = C_6 + 1 \text{ hub}$

$\chi(W_7) = 2 + 1 = 3$

Hub vertex will always require a different color.

$$\chi(W_n) = \begin{cases} 4, & \text{if } n = \text{even} \\ 3, & \text{if } n = \text{odd} \end{cases}$$

Q:- Find the Chromatic number of Complete bipartite graph $K_{m,n}$



$\therefore \chi(K_{m,n}) = 2$

i.e. One Color per Subset of Partition

Note:

① Complete bipartite graph can never be a NULL graph because no subset of partition can be Empty. and Each vertex of 1st non-Empty subset will Connect with each vertex of 2nd non-Empty subset. Hence at least one Edge will be present in the graph

But, a bipartite graph may be a NULL graph

eg.



it is a bipartite graph, And no Edge ∴ NULL graph

① For a Complete bipartite graph Chromatic no. is always = 2

② But for a bipartite graph Chromatic no. is 1 (when NULL graph) or 2 (otherwise)

Note: ① Chromatic no. of a bipartite graph is 1 or 2

i.e. $\chi(\text{Bipartite graph}) \leq 2$

② Bipartite graph are also known as bi-Colorable graph {i.e. 2-Colorable}, but not 2-chromatic

③ If Chromatic number of a graph 'G' is '2' then graph 'G' is a bipartite graph.

i.e. if $\chi(G) = 2$, then G is bipartite

Today's Topic

Note:

1. In a graph G, if there exists at least one cycle of odd length, then $\chi(G) \geq 3$

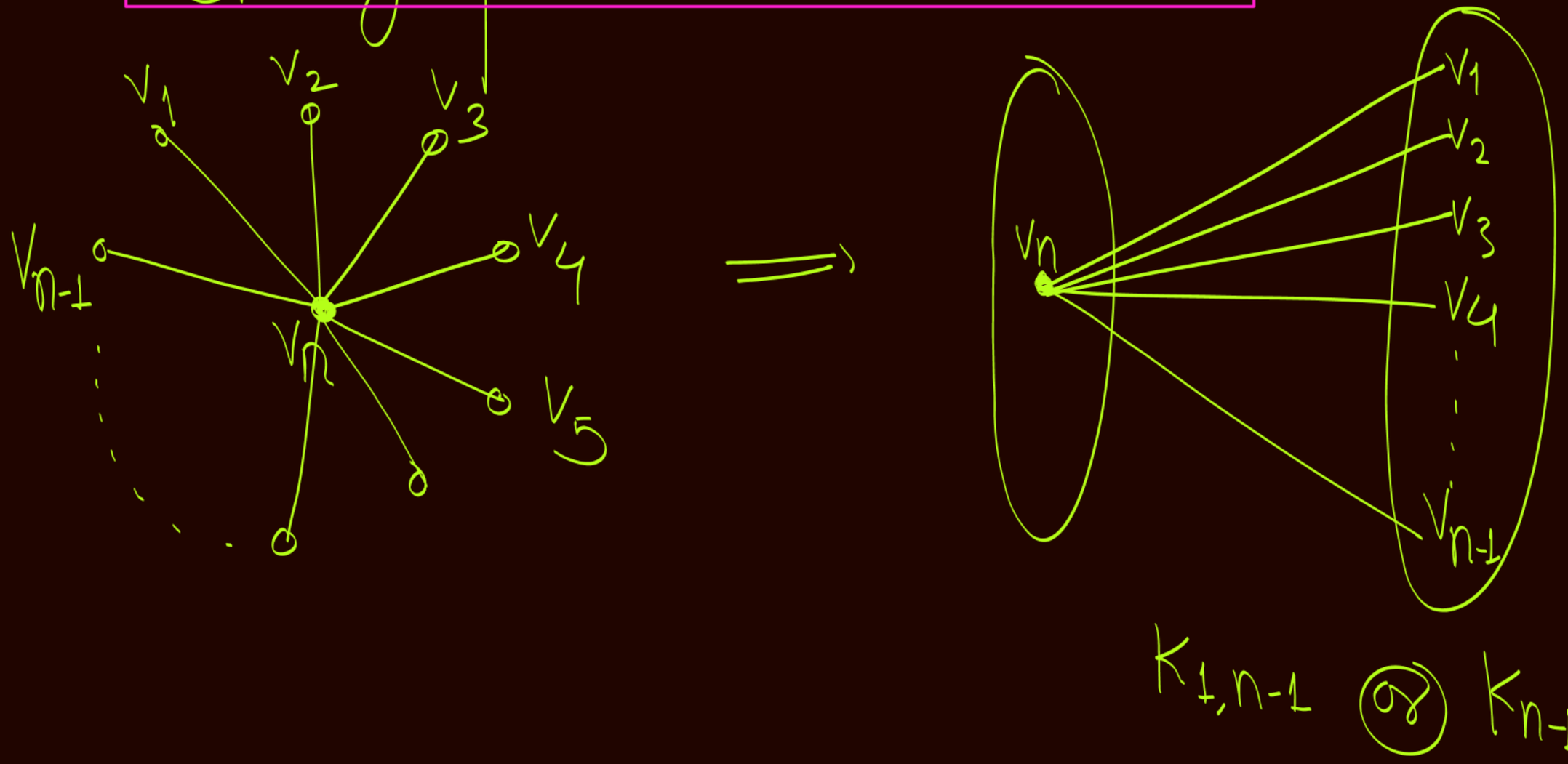
2. In a non-NULL graph G, if Every cycle is of Even length then $\chi(G) = 2$

3. In a graph G, if all cycles are of even length then graph G is a bipartite graph {Provided $|V| \geq 2$}

4. If $\chi(G) = 2$, then 'G' is a bipartite graph and hence Every cycle in graph G is of Even length

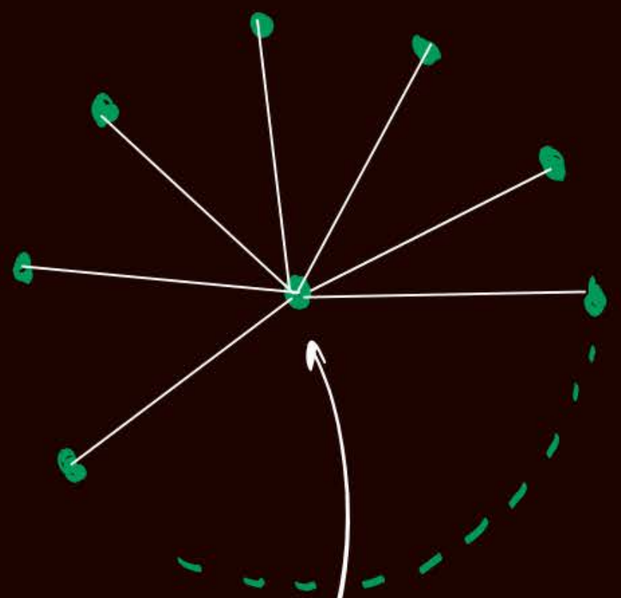
5. For a non-NULL tree (ie, for any tree with two or more vertices) the Chromatic number = 2

Star graph with n- vertices



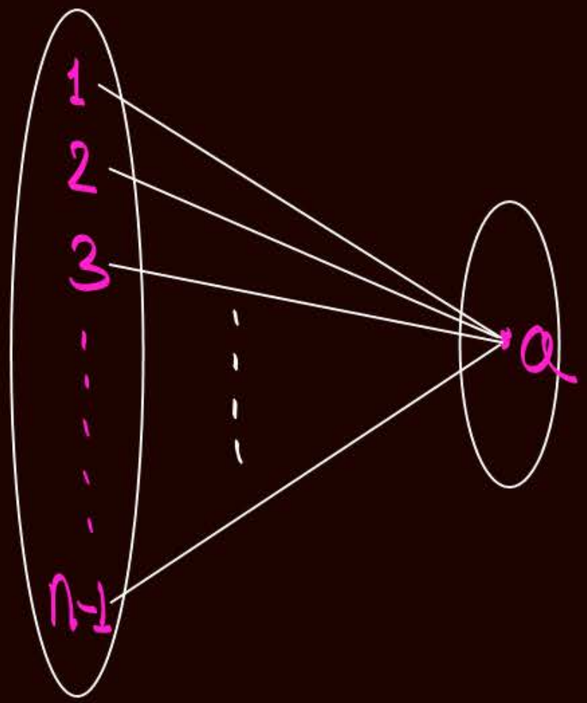
$K_{1,n-1}$ ⊗ $K_{n-1,1}$

Q:- Find the Chromatic number of a Star graph with 'n' vertices



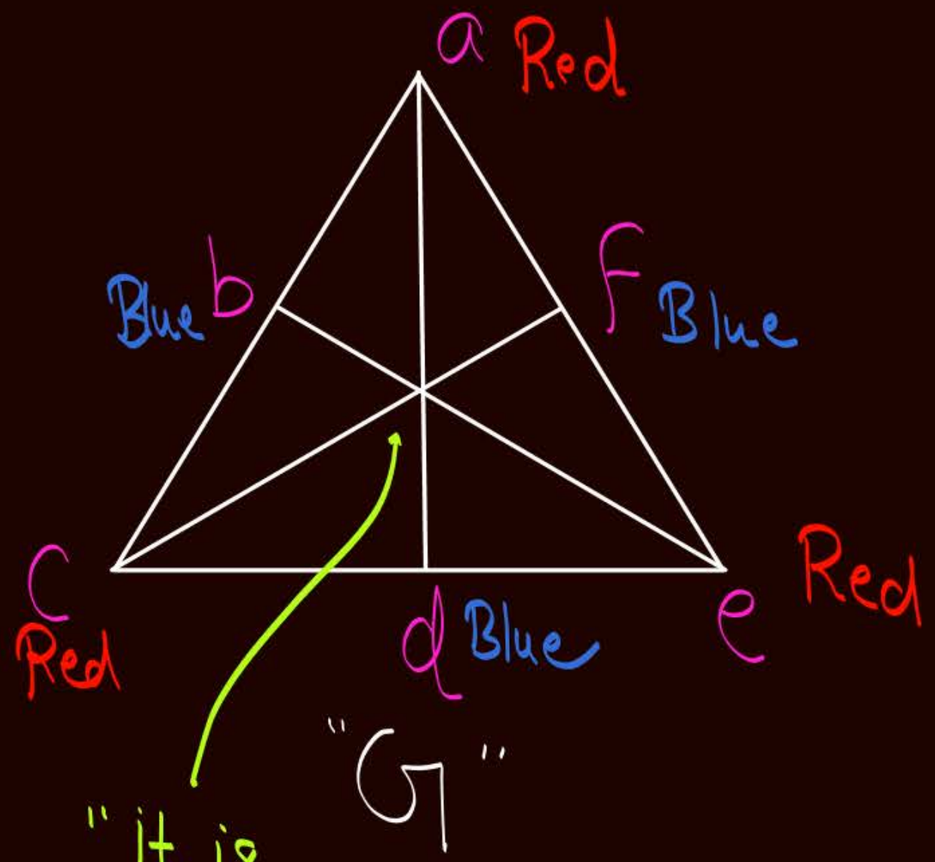
1 hub vertex
& (n-1) vertices at Periphery

⟹ A star graph with n-vertices can be visualized as $K_{1,n-1}$ (or) $K_{n-1,1}$



$\therefore \chi$ (Star graph with n-vertices) = 2

Eg:- Find the Chromatic number of following graph "G"



- We can vertex color the graph using two colors.

  $\therefore \chi(G) \leq 2$ —— eqⁿ ①

- Graph G is not a NULL graph

  $\therefore \chi(G) \geq 2$ —— eqⁿ ②

By eqⁿ ① & eqⁿ ②

$$\boxed{\chi(G) = 2}$$



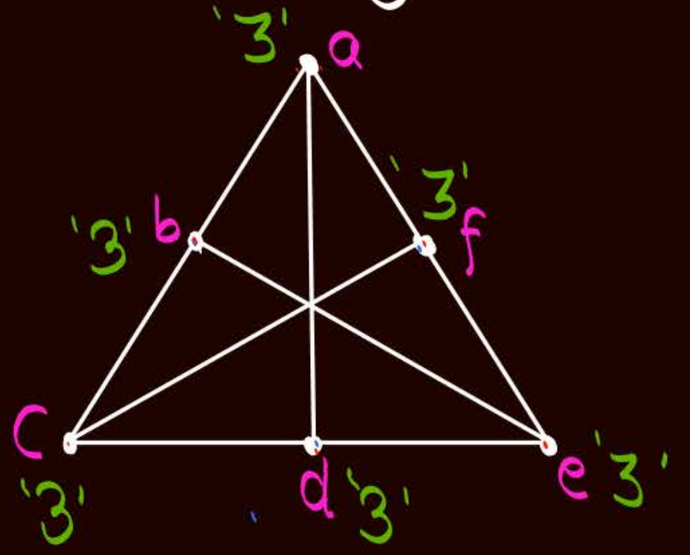
Topic : Welsh Powell's Algorithm

It is used to check the upper bound on chromatic number of a graph G.

1. Arrange the vertices of the graph G in non increasing order of their degrees.
2. If two or more vertices are having the same degree then use alphanumeric order.
3. Assign the colors to the vertices in non increasing order of their degrees such that no two adjacent vertices have the same color.
4. If Welsh Powell's algorithm uses "m" colors for a graph G, then $\chi(G) \leq m$. {i.e. graph G is m-Colorable}

'm' is the upper bound on Chromatic No.

eg:- Find the Chromatic number of following graph "G" using "Welsh Powell's algo"



"G"

All vertices are of same degree
∴ Use alphanumeric order

Vertices in non-increasing order of their degrees	a	b	c	d	e	f
Color Used	Red	— Green	Red	— Green	Red	— Green

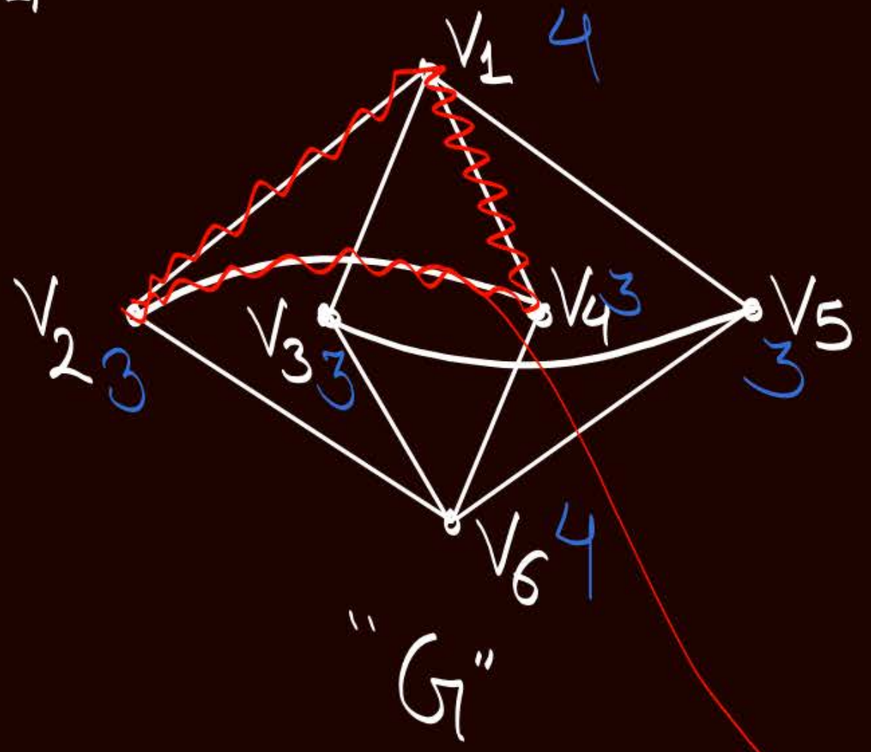
↳ Welsh Powell's algo uses '2' Colors
∴ $\chi(G) \leq 2$ ——— (1)

G is not a NULL graph
∴ $\chi(G) \geq 2$ ——— (2)

By eq^{n} (1) & eq^{n} (2)

$\chi(G) = 2$

Q:- Find the Chromatic number of the following graph 'G'



Vertices in non-increasing order of deg.	V_1 (deg=4)	V_6 (deg=4)	V_2 (deg=3)	V_3 (deg=3)	V_4 (deg=3)	V_5 (deg=3)
Colors	Red	Red	Green	Green	Blue	Blue

Welsh Powell's algo uses '3' colors
$\therefore \chi(G) \leq 3$ — ①

Odd length cycle exist in graph G, $\therefore \chi(G) \geq 3$ — ②

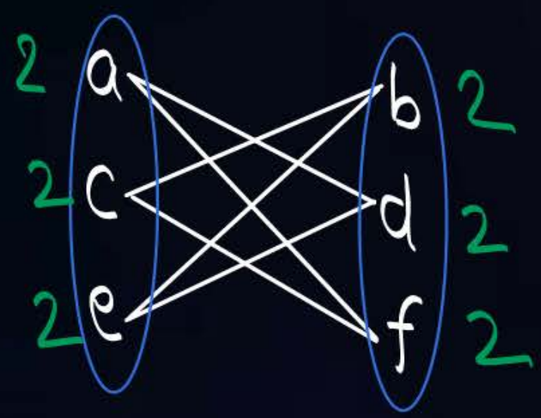
By ① & ②

$\chi(G) = 3$

Topic : Welsh Powell's Algorithm

- Welsh Powell's algo need not produce optimal result in all cases

eg: Find the chromatic number of following graph G



G

Same degree : Use alphanumeric order

Vertices in non-increasing order of deg	a	b	c	d	e	f
Color used	Red	Red	Green	Green	Blue	Blue

Given graph is a bipartite graph $\therefore \chi(G) = 2$

But Welsh Powell's algorithm uses '3' Colors.

Edge Coloring:- An assignment of Colors to the Edges of graph G, such that no two adjacent edges are Colored using the same color is called "Edge Coloring" of graph G.

* Minimum number of Colors required for Edge Coloring of graph G is Called "Index number" of graph G.

Matching



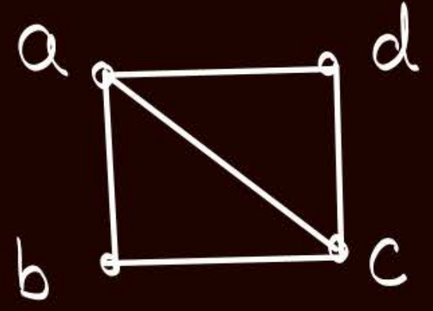
Topic : Matching

A subgraph $\acute{M}$ of graph $\acute{G}$ is called a matching of graph G, if all the vertices of graph G are incident with at most one edge in subgraph M,

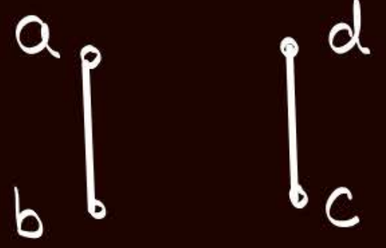
↳ All vertices of graph G must be present in sub-graph M

In a Matching of graph G, $\deg(v) \leq 1 \quad \forall v \in G$

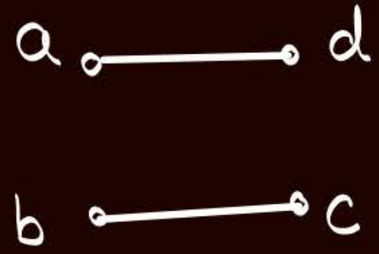
eg:



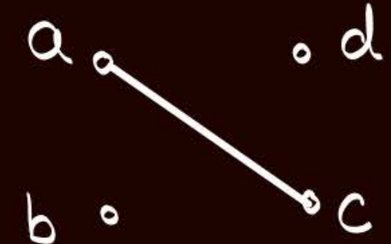
G



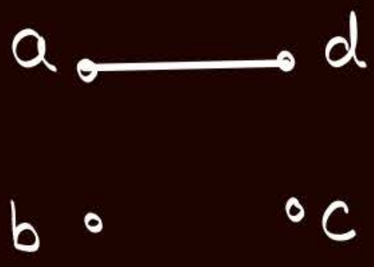
"M_1"



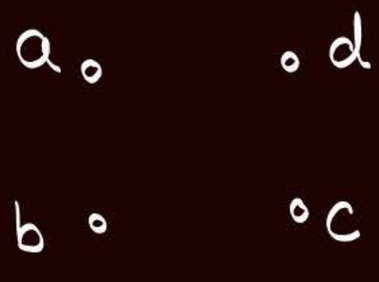
"M_2"



"M_3"



"M_4"



"M_5"

In a matching of graph G, adjacent edges are not allowed.

M_1, M_2, M_3, M_4, M_5 all are matching of graph G

Other matchings are also possible for graph G.

Note:

① In a Matching of graph G.
if $\deg(v) = 1$, then vertex 'v' is called a Matching Vertex

② In a Matching of graph G.
if $\deg(v) = 0$, then vertex 'v' is called a non-Matching Vertex

Maximal

↓

Nothing can be added, without destroying the Property

Minimal

↓

Nothing can be removed, without destroying the Property



Topic : Maximal Matching

- A matching of graph G is called a maximal matching, if no other edge of graph G can be added to that matching without destroying its property of being a matching.

In above example,

M_1, M_2 and M_3 are maximal matching of graph G.



Topic : Maximum Matching / Largest Maximal Matching

- The matchings of graph G with maximum number of edges are called maximum matching of graph G.

{ There may be more than one maximum matching for the given graph }

In the above example,

M_1 & M_2 are maximum matching

{ M_3 is maximal matching but not a maximum matching }

All maximum matching of graph G are also maximal matching of graph G, but all maximal matchings need not be maximum matching



Topic : Matching Number

* Matching Number of Graph G = Number of edges in any one of the maximum matching of graph G.

In the above example,

Matching no. of G = 2

↑

w.r.t. M_1 or M_2

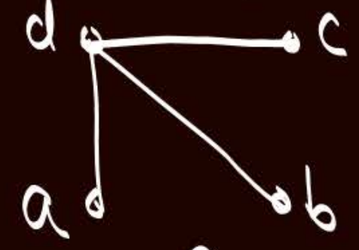
Note:-

Let G be a graph with n-vertices, then

matching number of graph G is $\leq \left\lfloor \frac{n}{2} \right\rfloor$

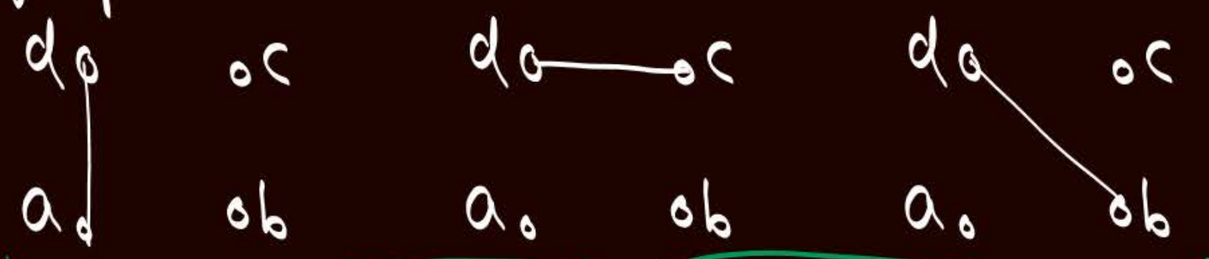
↑ it is the upper bound on matching No.

eg: Consider the following graph



G

in graph G, $n=4$



All are maximum matching of graph G

∴ Matching No. of $G = 1 < \left\lfloor \frac{4}{2} \right\rfloor$

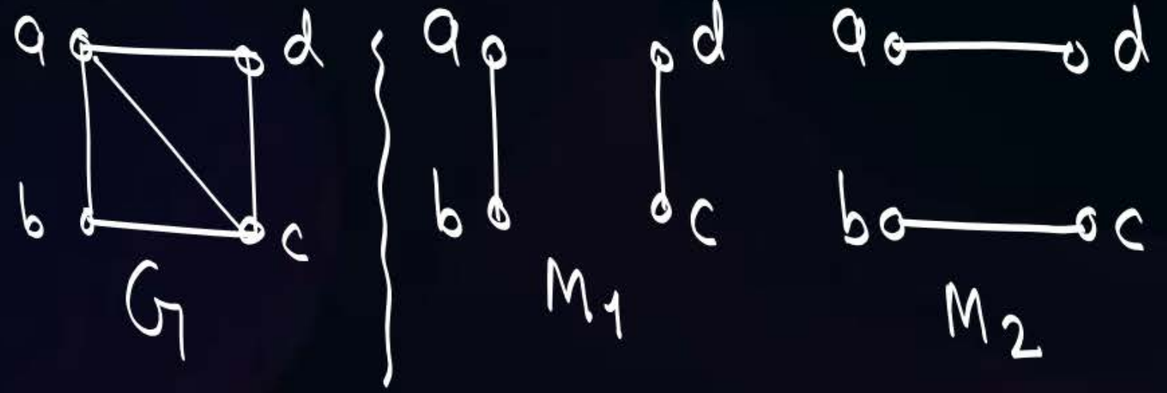


Topic : Perfect Matching

* A matching of graph G_1 is called a perfect matching if and only if degree of each vertex in the matching of graph G_1 is '1'.

in a perfect matching, $deg(v) = 1, \forall v \in G_1$

Eg:

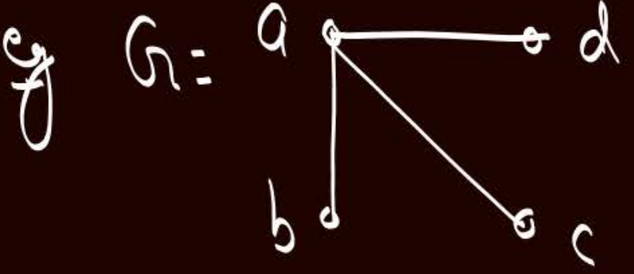


Both M_1 & M_2 are Perfect matching of graph G_1.

Note :-

① Every perfect matching of graph G, is also a maximum matching of graph G.

② But, Every maximum matching of graph G need not be a perfect matching

eg G: 

} Maximum matching exist for graph G, but no perfect matching

Note:-

If perfect matching exists for graph G, then number of vertices in graph G are even.

(Converse of the statement need not be true)

Note:-

Let G be a graph with n-vertices,

If perfect matching exists for graph G, then n = even.

and Matching Number of graph $G = \frac{n}{2}$

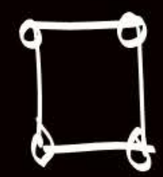
Note:-

① Perfect matching need not exist for Every graph

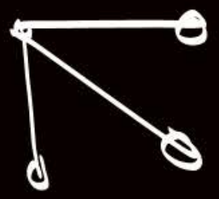
② In a graph G,
If no. of vertices are odd, then perfect matching
Can never exist for that graph

③ In a graph G,
If no. of vertices are even, then Perfect matching
may or may not exist for that graph

eg.

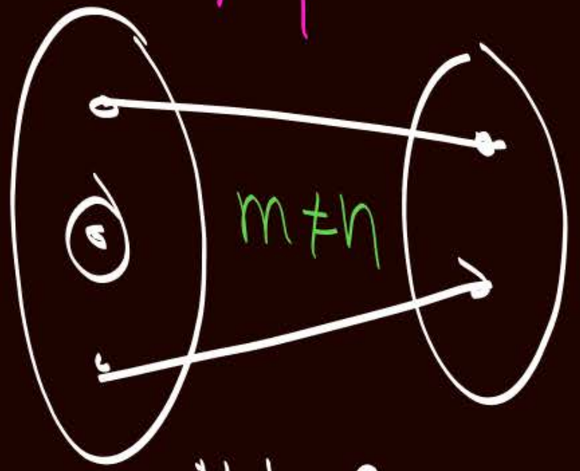


eg.

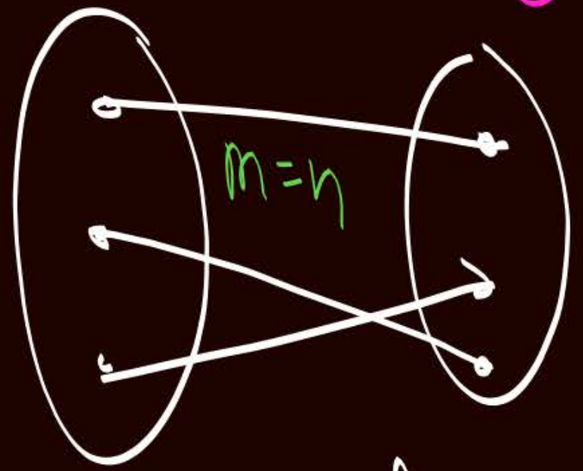


- For a Complete graph K_n Perfect matching exists if and only if n = Even in K_n

- In a Complete bipartite graph $K_{m,n}$ Perfect matching exist if and only if $m = n$

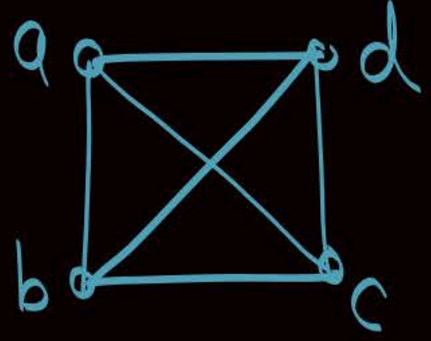


Not a Perfect matching

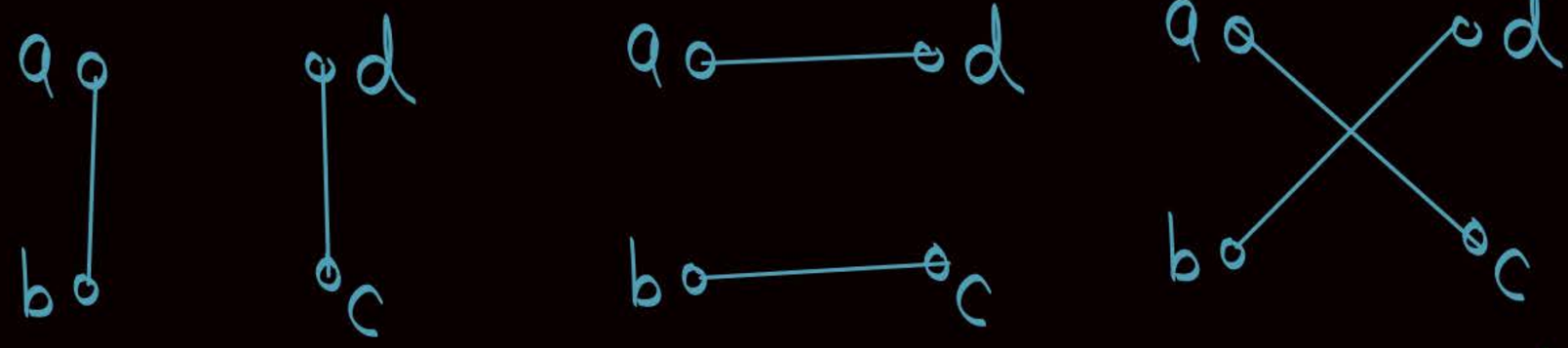


it is Perfect matching

* Let K_n is a Complete graph, where 'n' is Even.
Find the **no. of Perfect matching** in Complete graph K_n

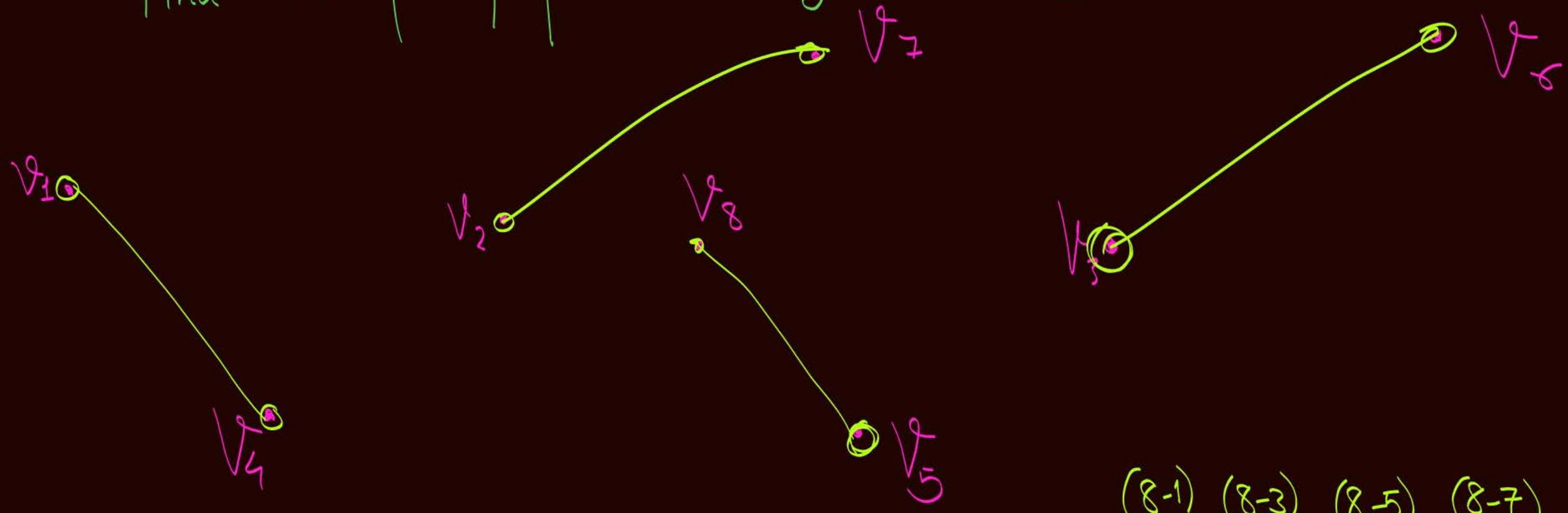


K_4



These are the only three Perfect Matchings Possible for K_4

Find no of perfect matchings in K_8



(8-1) (8-3) (8-5) (8-7)

no of perfect matchings in $K_8 = 7 * 5 * 3 * 1$

$= \underline{105}$

Number of Perfect matching in K_n {n = Even}

Number of Perfect matching in K_n

$$= (n-1)(n-3)(n-5)\ldots\ldots 5\cdot 3\cdot 1$$

$$= (n-1)(n-3)(n-5)\ldots\ldots 5\cdot 3\cdot 1 * \frac{n(n-2)(n-4)\ldots 6\cdot 4\cdot 2}{n(n-2)(n-4)\ldots 6\cdot 4\cdot 2}$$

$$= \frac{n(n-1)(n-2)(n-3)\ldots\ldots 6\cdot 5\cdot 4\cdot 3\cdot 2\cdot 1}{n\ (n-2)(n-4)\ldots 6\cdot 4\cdot 2}$$

$$= \frac{n!}{(2\cdot\frac{n}{2})\cdot(2\cdot(\frac{n}{2}-1))(2(\frac{n}{2}-2))\cdot(2\cdot 3)\cdot(2\cdot 2)\cdot(2\cdot 1)}$$

$$= \frac{n!}{(2)^{n/2}\cdot\{(\frac{n}{2})\cdot(\frac{n}{2}-1)\cdot(\frac{n}{2}-2)\ldots 3\cdot 2\cdot 1\}} = \boxed{\frac{n!}{(2)^{n/2}\cdot(\frac{n}{2})!}}$$

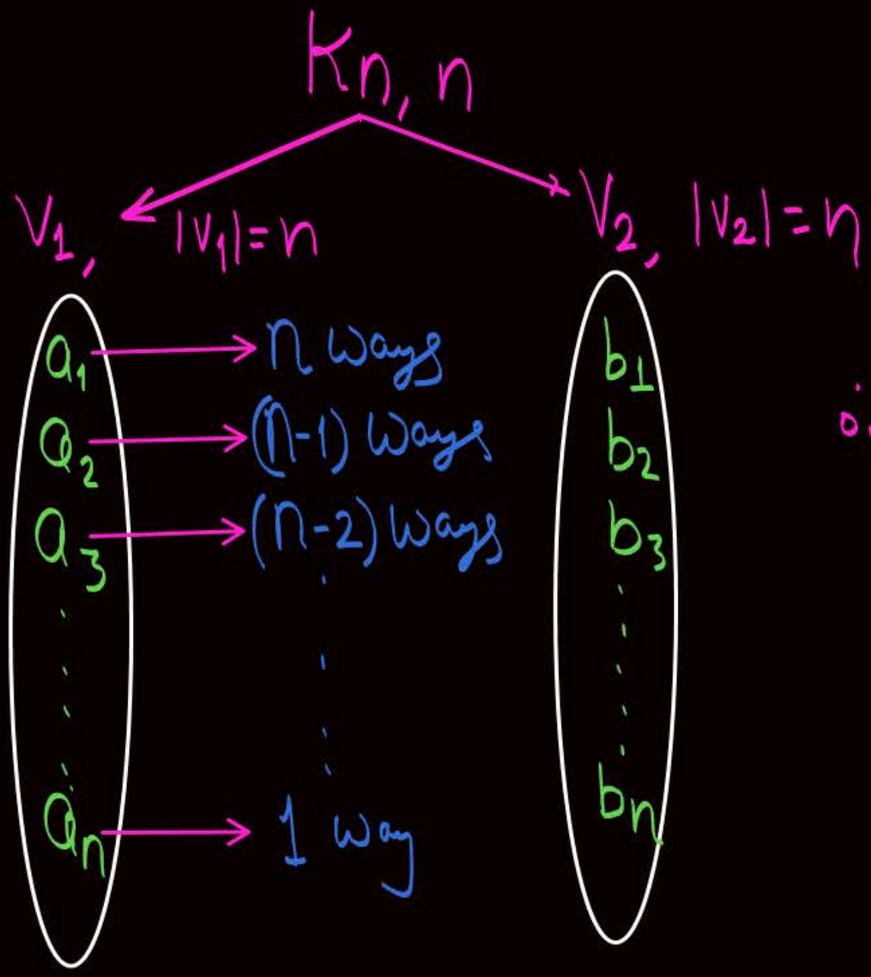
∴ Number of Perfect matching in K_{2n}

$2n$ — it will always be even

just replace 'n' by '2n'

$$\therefore \text{Number of Perfect matching in } K_{2n} = \frac{(2n)!}{(2)^{\frac{2n}{2}} \cdot \left(\frac{2n}{2}\right)!} = \boxed{\frac{(2n)!}{2^n \cdot n!}}$$

Q:- Find the number of perfect matching in Complete bi-partite graph $K_{n,n}$



∴ # Perfect matchings in Complete bipartite graph $K_{n,n}$ $= n(n-1)(n-2)\ldots 3 \cdot 2 \cdot 1$

$= n!$

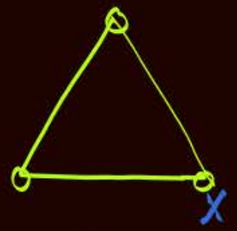
H.W

Q:- Find the matching number of Complete graph K_n



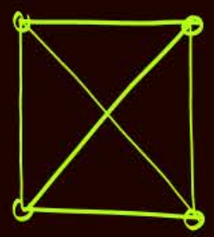
K_2

Matching No. of K_2 =



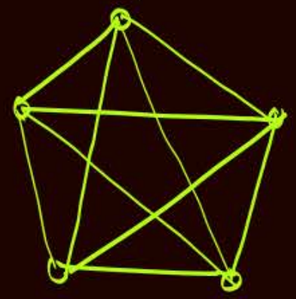
K_3

Matching No. of K_3 =



K_4

Matching No. of K_4 =

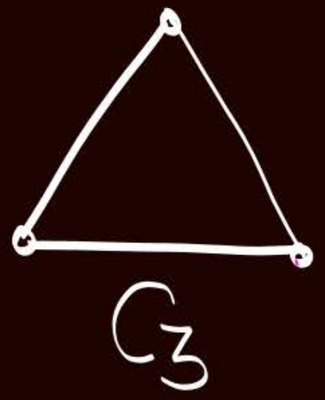


K_5

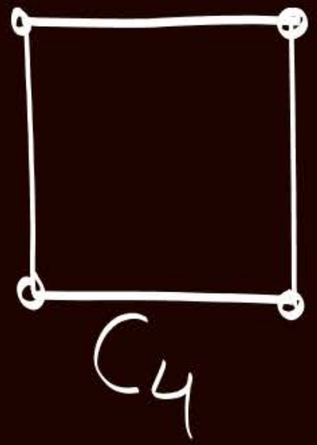
Matching No. of K_5 =

Q:. Find matching number of Cycle graph C_n

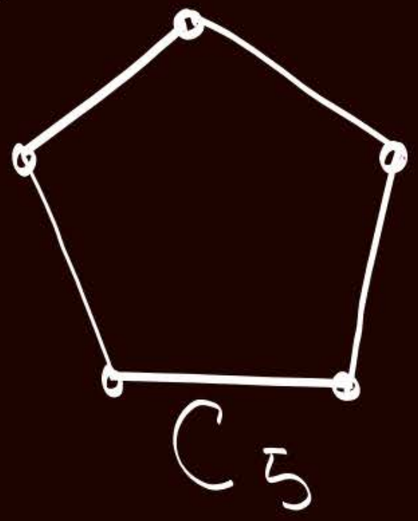
H.W.



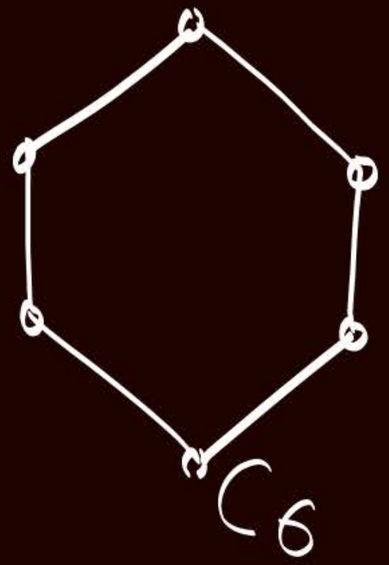
Matching No. =



Matching No. -



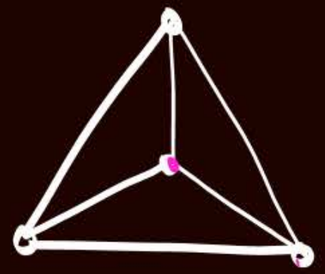
Matching No. =



Matching No. =

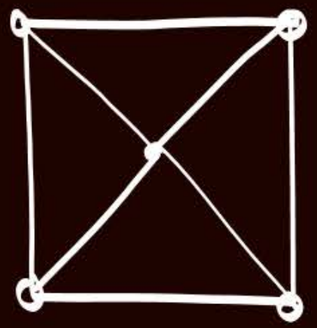
Q:. Find matching number of Wheel graph W_n

H.W.



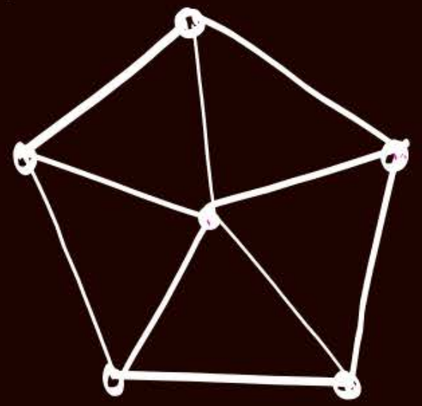
$W_4 = C_3 + 1$

Matching No. =



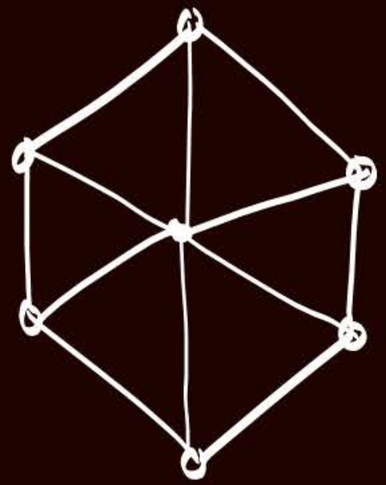
$W_5 = C_4 + 1$

Matching No. =



$W_6 = C_5 + 1$

Matching No. =



$W_7 = C_6 + 1$

Matching No. =

Q:- Find the matching number of Complete bipartite graph $K_{m,n}$

H.W

Q:- Find the matching number of Star graph with 'n' vertices.

HW



2 mins Summary

- Topic: Vertex coloring
- Topic: Chromatic number
- Topic: Edge coloring
- Topic: Matching

Slide

THANK - YOU

Slide